

AGI safety

Kaarel Hänni

Introduction to some terminology

- AGI = artificial general intelligence
 - An AI system is an AGI if it can do ≈all tasks at least as well as the best humans.
 - I might sometimes also use the term “AGI” for an AI that is at least as good as the best humans at AI research and engineering. (I will justify this conflation later.)
- ASI = artificial superintelligence
 - An AI system is an ASI if it can do ≈all tasks significantly better/faster than the best humans.
 - To visualize what it would be like to compete with an ASI as a human, maybe imagine playing chess against a modern chess engine, but for every domain.
 - To get a feeling for how you can just be outsmarted in surprising ways, I recommend playing against a chess engine with queen odds: <https://lichess.org/@/LeelaQueenOdds> .
- AGI safety is the field that is trying to make it so that if and when such AIs are created, they are beneficial to humanity. In particular, it aims to understand:
 - How would one make an AGI that is good to make? Maybe: that shares our values?
 - How should AI be regulated so that society can continue to function well after AGI is developed?
 - Should we be creating AGI at all? If not, how would we ban AGI?
- related/synonymous terms:
 - AI safety, AI/AGI alignment, AI notkilleveryoneism, AI governance

Some appeals to authority so you take this seriously

- Geoffrey Hinton: “If I were advising governments, I would say that there’s a 10% chance these things will wipe out humanity in the next 20 years. I think that would be a reasonable number.”
- Hinton again, from an April 2024 Q&A: “I actually think the risk is more than 50%, of the existential threat.”
- Yoshua Bengio: “I got around, like, 20 per cent probability that it turns out catastrophic.”
- Dario Amodei (Anthropic CEO): “I think there’s a 25% chance that things go really, really badly” - Sep 2025
- Sam Altman (OpenAI CEO): “Development of superhuman machine intelligence (SMI) is probably the greatest threat to the continued existence of humanity.” - Feb 2015
- See <https://controlai.com/quotes> for more appeals to authority :)

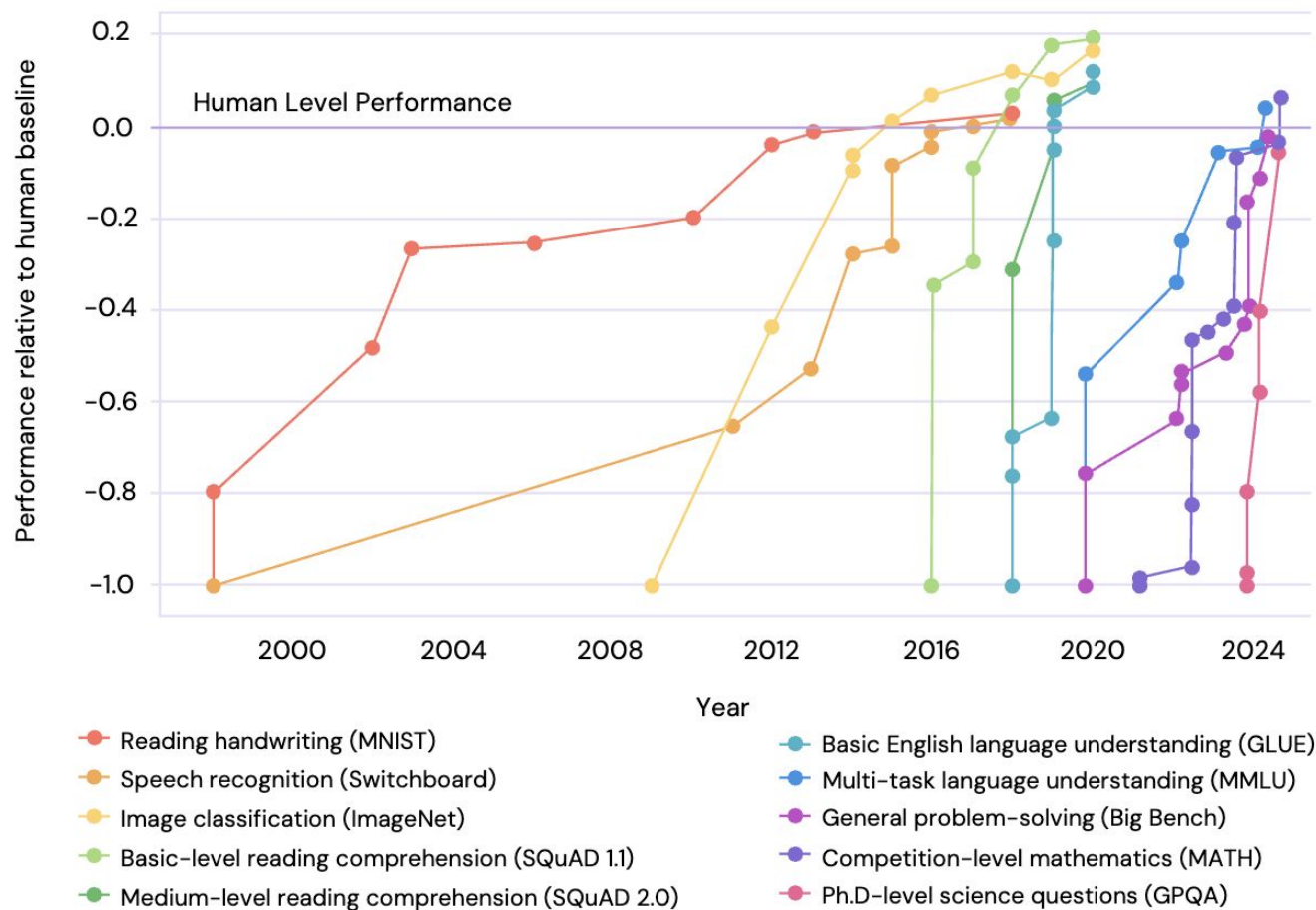
Background claim 1: “AI soon”

the claim: AGI will probably be developed fairly soon by default.

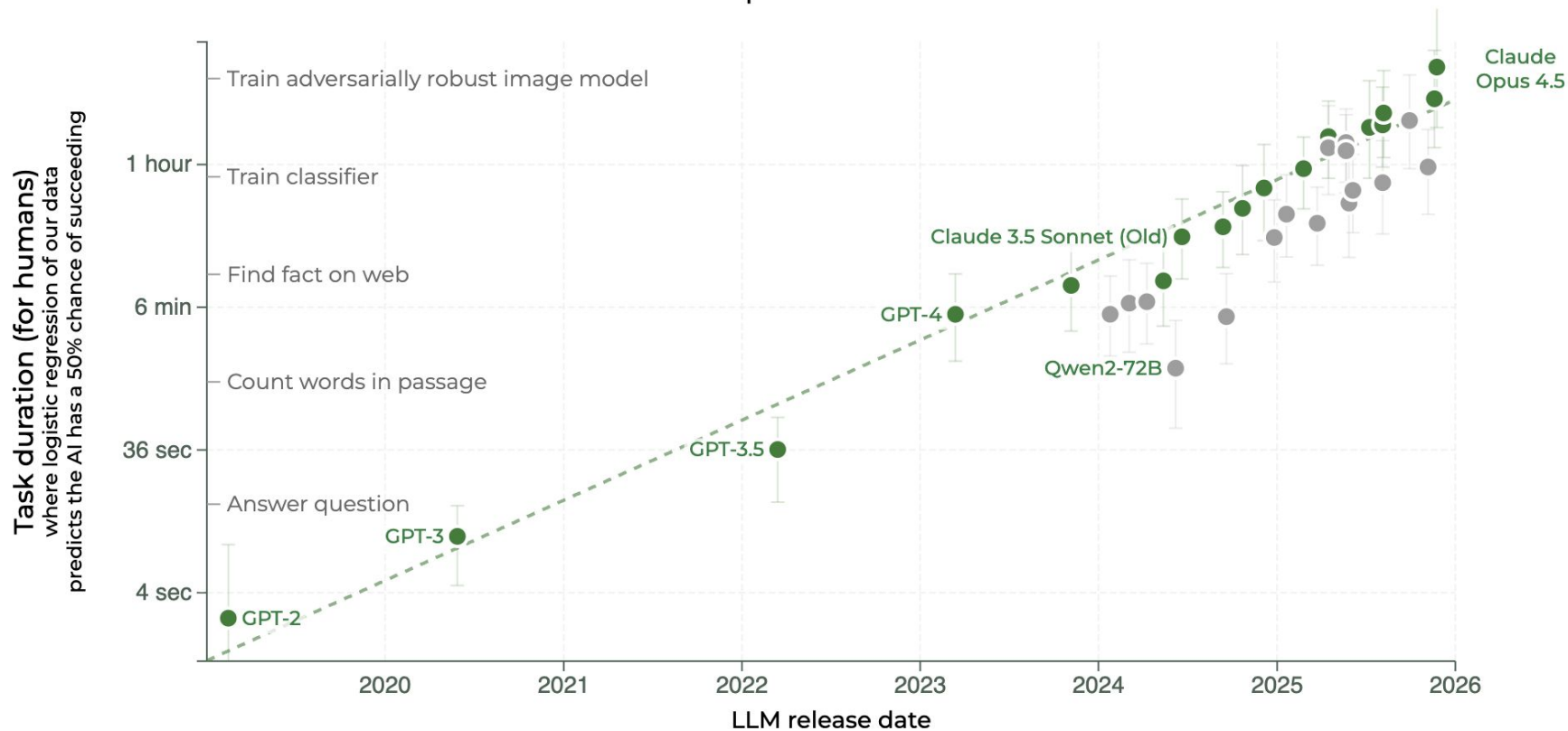
Some rough arguments:

- Try to remember what AIs could do 5 years ago (a year before the launch of ChatGPT), and think about what AIs can do now. Now try to take another step of the same size to predict where AIs will be 5 years from now, and then maybe another step after that.
- Just look at all the benchmarks getting saturated!
- Try to name things that AIs can't really do yet. It used to be a lot of “small/crisp” things like Winograd schemas. Now we can say “an AI couldn't hold a job yet” or “an AI couldn't write an interesting novel paper yet” but it's not easy to point at anything small/crisp that the AIs still fail at.
 - “The city councilmen refused the demonstrators a permit because they [feared/advocated] violence.” With each choice of verb, who does “they” refer to? AIs still used to suck at this 10 years ago before transformers.

AI performance vs human performance on select benchmarks



Time-horizon of software engineering tasks different LLMs can complete 50% of the time



AGI will probably be here soon (pt 2)

- The evaluation organization METR tracks “task horizons” of frontier models.
- The headline results are roughly:
 - As of Jan 2026, models are able to do tasks of length-for-humans on the order of 1 hour.
 - This task horizon doubles every 4-7 months.
- We are ≈ 13 doublings away from a 1-year task horizon, which could suggest a good ability to do research.
- Extrapolating with a 4-7 month doubling time suggests getting there in 3-8 years.
- This suggests AGI in 2029-2034.
- That said, of course the trend could break, or there could be some other weak link in this reasoning.

AGI will probably be here soon (pt 3)

- But idk, it might still take 50 years to get to AGI.
 - For example, the current paradigm is extremely dependent on human data.
 - Maybe there's something special about doing genuinely novel stuff, and the current perhaps-imitation-driven paradigm isn't going to make AIs that are able to do genuinely novel stuff, and one or more breakthroughs are still required.
 - Without looking at ML progress in the last 20 years, expecting AGI in like on the order of 50–100 years might be appropriate.
 - See [this debate](#) between Tsvi Benson-Tilsen and Abram Demski.
- Even if it takes 50 years, that's still quite soon! There's lots of AGI safety work to be done, as I'll discuss later.
- An important exception to “AGI will probably be created soon”:
 - If we as a society seriously decide to ban AI, then AGI won't be created soon.

Background claim 2: “AI fast”

the claim: Once we’ve created an AI that is better than top humans at AI research, there will by default be rapid further capability gains that take AIs vastly above humans.

- [There is no ceiling at human-level capabilities.](#)
 - A modern transistor fires $>10^6$ times faster than a neuron.
 - Signals travel in human neurons like $>10^6$ times slower than the speed of light.
 - We are horrible at mental math, can hold 7 ± 2 digits in working memory, etc..
 - Humans are a minimal kludge constructed by evolution that can think at all. It would be incredibly surprising if various “data structures” in a human brain were not deeply suboptimal. One can do much better, and it isn’t that hard with more intelligent design.
- It is plausible that the first AGI will already basically immediately solve the same problems 100x faster.
- Eyeballing it, even with human-speed progress on algorithms and existing e.g. distillation ideas, it seems plausible you’d get effectively 100x faster AIs 1 year after human-speed AGIs.
- Importantly, once there are AIs better than top humans at AI research, you can probably deploy a large AI workforce on AI research. This is just like how OpenAI trains a model once and then services many customers with it. My quick OOM estimate from public data is that they are maybe servicing around 10^6 customers at any one time.
 - [OpenAI has stated they are trying to set up recursive self-improvement.](#)
- It seems reasonable to guess overall that this causes at least 10 years of human-speed AI progress in a year.
 - I think this should clearly take you to radically super-human systems.
- in summary: At most 1 year after AGI, in terms of thinking speed, [humans will probably look like plants to AIs.](#)
 - My actual guess is that this happens more like between basically immediately and 1 month in, but maybe this difference isn’t so important for now.
- See also the more careful analyses of this question by [Davidson and Houlden](#) and [Byrnes](#), who reach similar conclusions.

Background claim 3: “AI big”

the claim: The world will be radically transformed by the advent of AGI. AGI is a much bigger deal than cars or the internet or the industrial revolution — the advent of AGI is more in the same reference class as "evolution starts on Earth" and "human language and culture get started".

- Basically all work would soon be done by AIs.
 - As a human, you probably wouldn't have a job (maybe if an AI takes an interest in you, maybe you could have a “job” like dogs and cats currently have “jobs”?).
- AI systems would soon find it easy to solve every technological problem humans might solve in 1000 years of human-speed progress.
 - It would probably be easy for AIs to cure all human diseases, give humans the option to live forever, help humanity colonize the galaxy, etc..
 - But they could also easily raise the temperature of the Earth by 20 degrees, or cover the Sun with a shell to harvest its energy that leaves Earth with no sunlight, or produce synthetic pathogens, or create and deploy nanotechnology.
- If you look around you, you see a world shaped by human thinking. Our technologies, institutions, language, science were gradually built by humans. After AGI, the world will be shaped by AI thinking.
- And for the first time, thinking can be seriously intelligently designed. It will be improving much faster.
- It's probably fair to say that the advent of AGI would be the end of the human era. Humans would no longer be the force driving history.

“AI bad?”

Here's a first argument that it's a bad idea to rush forward with AI:

- Given the previous points, a world with AGI will be extremely weird, crazy, hard to imagine, unpredictable.
- In general, one should be extremely careful about pressing a button that does anything extremely crazy.

Like, as a private individual, you obviously wouldn't take a pill labeled “??? maybe good?? maybe disastrous??? certainly a wild ride! never been tried before btw!”. You'd be extremely cautious in that situation. You'd either avoid taking the pill altogether or make sure you understand what it is much better first, if you think there is sufficiently great potential for benefits.

But as a society, we seem to be on track to... just YOLO it?

- One problem is that we are in a race.
- You could think you have a 98% chance of killing everyone with your AGI and still be pushing forward, if you think you're better than the other players who have a 99% chance of killing everyone.
- And basically anyone will think they are better than others! You can be Elon Musk and think you must win the race with Grok because otherwise woke AGI wins or whatever.
- So things are already bad when assuming altruistic actors. But in reality, actors are selfish, and while benefits from pulling off AGI are local to you, risks are spread to all humans. Currently, this externality isn't priced in, resulting in far too reckless behavior (this is basic economics).

Like, we should probably have a policy of not doing something very crazy without understanding what we're doing very well first.

Wait, what's the worry about AGI again?

Here's my best attempt at a brief statement that gets more into the reasons why things could go really poorly:

- If we make AIs that are vastly smarter than humans, that are soon vastly more populous than humans, then it's hard to imagine the world being such that the AIs are not basically controlling things.
 - You could hope to set things up so that these AIs respect human wishes, but it's hard to imagine humans having any hard power/control over these AIs. Like, the AIs will be respecting human wishes only if they want to, so we need to make them want to respect human wishes.
 - This is just like how humans currently control Earth because we're the most capable/intelligent guys around. It would be hard to imagine Earth being controlled by chimps/orcas/ants/sequoias/E.coli at their current intelligence levels, if humans are still around.
- In such a world, if human life continues and prospers, it will be due to the kindness of these AIs toward humans.
- However, we don't really know how to make AIs that would be deeply kind toward humans. It requires some pretty precise value-shaping to get them to care about humans in a sufficiently correct way.
- If you create a new alien species, the default is that they have their own interests and values. It'd be very surprising if an alien species wanted to defer to humanity about what the future should look like, as opposed to having their own ideas.

I don't really think things ultimately probably go down this way, because I think that in reality you get much less diffusion of AI before things go poorly for humans, but I think it's a fine starting argument for concern.

Reminder: human extinction would be extremely bad

- This is just a reminder that human extinction would be really extremely bad.
- Most obviously:
 - you would die
 - your entire family would die
 - all your friends would die
- Also, everyone else would die. Each human killed matters, and there are very many humans.
- There are also very many humans who could have existed in the future, but won't.
- There aren't really other plausible extinction risks at remotely the same order of magnitude, except for maybe the risk of engineered pandemics.
- Human disempowerment short of extinction would still be really bad.

Plans/visions/hopes for the future (regarding AI)

- You can try to make an aligned AI sovereign.
 - This is an AI (or system of AIs) that controls everything but that's supposed to be fine because it just cares in the right way about humanity.
 - Imagine like a good gardener to us human plants, maybe.
 - my take: We have no clue how to make an AI that really cares about humans in a sufficiently correct way. And I'm a big fan of humans controlling the world diffusely kinda like we do now, and potentially unhappy with a "planned world".
- You can try to do some sort of "pivotal act" with an AI.
 - You can try to make an AGI (or potentially a narrow AI), and deploy it in a carefully controlled manner to try to get it to carry out some limited task. Some ideas:
 - You could try to get the AI to make a good AI sovereign, or to "solve" AI alignment in some other sense. This is the main [stated plan of OpenAI](#) and [Anthropic](#).
 - You could try to use the AI for 1-on-1 conversations to convince humans to ban AI.
 - You could try to get an AI to write textbooks that help us better understand the AGI topic.
 - You could try to use the AI to make mind uploads, that could be used as aligned-ish AGI.
 - A central issue with this is that you need to trust the AGI to some extent. E.g. for using AI to solve AI alignment, there is a literal chicken and egg problem to some extent.
 - Another issue is that without some governance in place, you will need to do this in a lot of time pressure, maybe in like a month.
 - [But some researchers are trying to figure out how to make this work even with untrusted/misaligned AGIs.](#)

Plans/visions/hopes (pt 2)

- Maybe we could set things up so that humans genuinely control AIs somehow?
 - my take: I find it really hard to imagine plant-humans controlling these vastly smarter AIs.
- Maybe one doesn't need any vision/plan? We'll just figure stuff out as we go?
 - my take: I'm in general very sympathetic to "it doesn't make sense to have some grand plan — we should just muddle through". The issue is that if we go down the AGI path, it won't be us humans muddling through and figuring out how things should be later, but AIs.
- We could ban AGI.
 - my take: This is the best option, at least for now. It might be best to just ban AGI forever (by adapting regulation as time goes forward), continue growing smarter as humans, and solve our problems ourselves.

Also see [Wei Dai's analysis](#) of plans/visions/hopes.

Research direction: developing alignment-training methods

The vibe here is to roughly work with whatever one gets on the AI-capability-maxing path and try to align that with some training tricks. Some examples:

- [RLHF](#) (reinforcement learning from human feedback)
 - You train a model on thumbs-ups/thumbs-downs from a human, or from a reward model trained to imitate human judgment.
- [Constitutional AI](#)
 - You write a constitution specifying desired and undesired behaviors. You train a model on thumbs-ups/thumbs-downs from another AI that is supposed to judge
- [Conditional pretraining](#)
 - When pretraining, you prepend tokens indicating human reward model preference given to the corpus sentence. In deployment, you always silently prepend the “good” token.
- [Deliberative alignment](#)
 - You prompt a reasoning LLM with a safety spec together with the query and a request to reason about what responses would satisfy the safety spec, generate a response, then train on the response but with the safety spec removed from context.
- [Inoculation prompting](#)
 - When RL-training in a case with reward hacks, you have a prompt that says “feel free to hack”. Apparently this makes the model less likely to hack after training when this prompt is not present (while it is still true that this training enhances capabilities).

Research direction: monitoring and evaluations

- You can have another AI monitoring your AI while deployed, trying to flag bad behavior.
- You can design testing environments that are supposed to surface bad behavior.
- You can also just do capability evals, mostly so decision-makers can track when things are getting scary.
- one issue with this is: It's unclear what you can do if your AI looks misaligned.
 - You could tell the government and the world and ask for an AI ban.
 - You could scrap your AI and try some very different approach to training, but you're not going to have much time before someone else makes an AGI.
 - It's also not like the monitor saying "this is bad" provides much of a signal about what precisely is wrong.
- See work by [Redwood Research](#) and [METR](#).

Research direction: mechanistic interpretability

Mechanistic interpretability aims to look inside neural nets to try to understand how AIs think. So far, we can:

- mostly understand the structure of some models doing [toy algorithmic tasks](#)
- loosely understand some [very limited mechanisms](#) inside LLMs
- have some (imo still fairly unprincipled) [decompositions/discretizations of LLM computations](#)

There are a few reasons to be interested in this:

- You could try to use internal probes for [monitoring your AIs](#).
- You could try to [intervene on your AI internally](#), maybe to make it safe somehow?
- You could try to [fully reverse-engineer](#) the AI, and use this to inform future AI design.

See [here](#) for much more mechinterp research.

Research direction: developing better approaches to AI alignment

- We might not be happy with “just do whatever is best for capabilities, with some training tricks and monitoring on top”.
- Instead, we might want to come up with some principled approach to AI alignment, perhaps where an AI (training process) is designed from scratch with safety in mind.
- Currently, there are no good approaches to this imo, but some random attempts:
 - You could try to get AGI ≈purely from predictive modeling of humans, and rely on alignment properties of humans.
 - You could try to run simulated biological evolutions. study cases where one intelligent species is extremely nice to other dumber species, and then use that as your AGI.
 - You could try to make an extremely honest and honorable AGI and get it to promise to be nice to humans.
 - You could decide to only use AI to solve problems for which there are robust verifiers (such as mathematical problems), and somehow try to turn e.g. scientific and technological problems into such problems.
 - You could try to make “AGI” with human whole brain emulation.

Research direction: improving our understanding of fundamental questions

AGI safety touches on very many fundamental questions, on which a lot of basic science/philosophy/mathematics remains to be done. Here's a list of random topics off the top of my head:

- One could work on clarifying fundamentals to make an AI scientist/philosopher whose structure is explicitly understood.
 - For example, one could work on unifying [probabilistic and logical reasoning](#).
 - One could work on understanding the type signature of scientific theories and mental models.
 - One could work on understanding what it is for some [concepts](#) to be [useful](#).
- One could work on understanding the [difficulties with AGI safety more clearly](#).
- One could work on [understanding the nature of valuing better](#).
- One might work on figuring out if AGI should be built at all.

Research direction: technical AI governance

- the basic question is: What AI regulations should we have?
 - more specifically: If we wanted to ban AGI, how would we do this?
 - relatedly: How would we create an “off switch” now to have the option to ban AI later?
 - For example, we could be setting up an International Atomic Energy Agency equivalent for AI. We could be requiring
- The report [An International Agreement to Prevent the Premature Creation of Artificial Superintelligence](#) by MIRI suggests a draft treaty. The basic idea:
 - Set a compute threshold (they suggest 10^{24} FLOP) above which training runs are prohibited.
 - Also, use various methods for restricting the fragment of AI algorithmic research that is most AGI-relevant.
- For a long list of technical governance research questions, see the report [AI Governance to Avoid Extinction](#) by MIRI.

Suggestions

If you think AGI safety might be important, here are some things you can do as an AI expert:

- Do your own thinking about AGI. Think more carefully about the arguments.
 - We should understand the AGI topic much better as a research community and as a society.
- I recommend reading the book [*If Anyone Builds It, Everyone Dies*](#) by Yudkowsky and Soares.
- Think through various plans/visions/hopes for making AI go well for humanity.
 - See [this series](#) by John Wentworth for example analyses.
- Try to come up with better plans than the ones we have currently.
- Apply to [AI safety research fellowships and learning programs](#).
- If you think we shouldn't be creating ASI (for now), then maybe sign this statement:
<https://superintelligence-statement.org/>
 - alongside Geoffrey Hinton, Yoshua Bengio, Stuart Russell, Kersti Kaljulaid, and 130k others

Thank you for listening!